

EDUVISION DV

Higher Education Analytics Dashboard

FINAL PROJECT DOCUMENTATION

1. Executive Overview

EduVision DV is a Tableau-based higher education analytics solution containing four connected dashboards: University Overview, Research Analytics, Student Analytics and Country Comparison. The submitted workbook contains multiple embedded CSV extracts supporting university-level, student-level and country-level analysis.

The central university dataset contains 204 records and 37 fields, with a 2024 ranking reference. The workbook also contains dedicated student, country and regional extracts. The report records the workbook's actual structure, measures and validation observations without introducing unsupported values.

Workbook item	Verified value
Analytical dashboards	4
University records	204
Ranking reference	2024
Country-ranking records	68
Regional trend records	4,486

Key KPI values calculated from the embedded workbook data:

Measure	Value
Average Overall Score	52.83
Average Academic Reputation	51.30
Average Citations per Faculty	58.01
Average International Research Network	50.05
Average Employment Outcomes	52.92
Average International Students	11.18%
Average Faculty-to-Student Ratio	20.77
Total Students	36,240,550

2. Project Scope and Objectives

- Compare universities using overall score, ranking position and academic indicators.
- Analyse research-related indicators including citations, international research networking and employment outcomes.
- Examine international student participation, faculty-to-student ratio, enrolment and student distribution.
- Compare countries using overall score, ranking position and tertiary-enrolment indicators.
- Observe regional education trends from the available regional extract.
- Provide connected dashboard navigation and interactive analytical views.

3. Data Sources and Workbook Architecture

The workbook is not based on one single flat table. Separate embedded CSV extracts support different analytical areas. The main university extract is used for University Overview and Research Analytics, while dedicated extracts support Student Analytics and Country Comparison.

Embedded dataset	Size	Purpose
EduVision_M1_Data.csv	204 rows × 37 columns	University ranking and institutional indicators
student_enrollment.csv	111 rows × 4 columns	Student enrolment
student_distribution.csv	111 rows × 3 columns	Student distribution
student_faculty.csv	243 rows × 3 columns	Faculty-to-student indicators
student_diversity.csv	243 rows × 4 columns	Student diversity
student_international.csv	243 rows × 3 columns	International student indicators
country_ranking.csv	68 rows × 4 columns	Country ranking indicators
top_countries.csv	68 rows × 3 columns	Top country comparison
education_benchmark.csv	35 rows × 6 columns	Education benchmark indicators
regional_trends.csv	4,486 rows × 4 columns	Regional education trends

4. KPI Framework

The workbook contains dedicated KPI worksheets for university, research, student and country measures. Key measures include:

KPI / measure	Calculated value
Average Overall Score	52.83
Average Academic Reputation	51.30
Average Citations per Faculty	58.01
Average International Research Network	50.05
Average Employment Outcomes	52.92
Average International Students	11.18%
Average Faculty-to-Student Ratio	20.77
Total Students	36,240,550
Country Average Overall Score	55.37
Country Average Rank	178.51
Average Tertiary Enrolment	58.02%

Interpretation should follow the field definitions in the workbook. In particular, the faculty-to-student measure is a numeric ratio and should not be treated as a direct measure of teaching quality.

5. Dashboard Structure

Dashboard	Analytical focus
University Overview	University ranking, overall score, academic reputation, distribution and institutional comparison.
Research Analytics	Research, citation, international research network and employment-related indicators.
Student Analytics	International students, faculty-to-student ratio, student distribution and enrolment measures.
Country Comparison	Country ranking, tertiary enrolment and regional education indicators.

6. Filters, Navigation and Interactivity

The workbook contains navigation and dashboard-action definitions connecting the four analytical dashboards. University, country and related selections were tested through Tableau actions, and the configured interactions responded as expected. The packaged workbook contains 22 Tableau actions.

7. Testing and Quality Assurance

Quality assurance was separated into data/structure checks and visual/runtime checks. Formatting consistency has been confirmed. The two target display resolutions for the visual QA check are 1366×768 and 1920×1080.

Test area	Status
KPI values against embedded data	PASS
Ranking/data consistency	PASS
Dashboard structure and worksheet presence	PASS
Configured Tableau actions	PASS
Formatting consistency	PASS
1366×768 runtime layout	PASS
1920×1080 runtime layout	PASS
Interactive filter/action response	PASS
Tooltip behaviour	PASS

TC-20 — Formatting and Visual Consistency: PASS. Dashboard formatting was reviewed for alignment, readability, number formats, filters, chart placement and consistent presentation.

8. Analytical Findings

- The university ranking extract contains 204 institutions. The average overall score is 52.83.
- The average academic reputation score is 51.30 and the average citations-per-faculty score is 58.01.
- The average international student percentage in the student-related data is 11.18%, while the average faculty-to-student ratio is 20.77.
- The student distribution extract represents 36,240,550 students in total.
- The country-ranking extract contains 68 records, with an average overall score of 55.37 and average rank position of 178.51.
- Average tertiary enrolment represented in the country benchmark data is 58.02%.

9. Data Quality and Limitations

Across the embedded CSV extracts, overall cell completeness was calculated at approximately 98.29%. The main university extract itself contains missing values in some secondary fields, although the five principal ranking KPI fields used for the core averages were complete in the inspected extract.

Some regional education-expenditure values require source-level verification before being interpreted as real-world economic measurements. The workbook should therefore be treated as a visualization of the supplied datasets, with source definitions and unusual values checked before external publication.

10. Technical Implementation and Delivery

Component	Implementation
Visualization	Tableau
Workbook format	Packaged Tableau workbook (.twbx)
Data format	Embedded CSV extracts
Dashboard count	4
Workbook actions	22 configured Tableau actions

11. Conclusion

EduVision DV provides a compact four-dashboard analytical environment for examining university performance, research indicators, student characteristics and country-level education measures. The report documents the actual workbook structure and calculated values, while clearly separating verified workbook facts from runtime checks that must be confirmed in Tableau.

Appendix: Core Workbook Inventory

- University Overview — KPI cards, ranking views, distribution and academic/institutional comparison.
- Research Analytics — research, citation, network and employment-related views and KPIs.
- Student Analytics — international students, faculty ratio, diversity, enrolment and distribution views.
- Country Comparison — country ranking, benchmark, top-country and regional trend views.